

An Observational Cross-Country Regression of Growth, Anthropometric Measurement and Nuclear-Armed Status

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Abstract

This paper estimates a cross-country observational regression of the form

$$y_i = a + bX_{1i} + c\mathbf{1}_{\{X_{2i}=1\}} + \varepsilon_i,$$

where y_i is the GDP growth rate (obtained from the IMF), X_{1i} is average erect penis length in centimeters (obtained from WPR), and X_{2i} is an indicator for whether country i is a nuclear-armed nation (obtained from SIPRI). The estimated ordinary least squares model is

$$\hat{y}_i = 7.2155 - 0.3541X_{1i} + 0.3876\mathbf{1}_{\{X_{2i}=1\}}.$$

The coefficient on X_1 is negative and statistically significant in this dataset, while the coefficient on nuclear-armed status is positive but statistically insignificant. The analysis is strictly associational and should not be interpreted causally.

The paper ends with “The End”

1 Introduction

Cross-country regressions are frequently used in economics, political science, international relations, finance, and security studies to explore empirical regularities across sovereign states. This paper studies the reduced-form relation between a dependent economic variable y , an anthropometric covariate X_1 , and a geopolitical-security indicator X_2 .

The estimated equation is

$$y_i = a + bX_{1i} + cD_i + \varepsilon_i, \tag{1}$$

where

$$D_i = \mathbf{1}_{\{X_{2i}=1\}}.$$

Since $X_2 \in \{0, 1\}$, the indicator variable is numerically identical to X_2 :

$$\mathbf{1}_{\{X_{2i}=1\}} = X_{2i}.$$

2 Data

The dataset contains $n = 43$ countries. The variables are defined as follows:

y GDP growth rate.

X_1 Average erect penis length in centimeters.

X_2 Nuclear-armed nation indicator, equal to 1 if nuclear-armed and 0 otherwise.

Table 1: Regression variables

Code	Country	y	X_1	X_2
AZ	Azerbaijan	3.2	13.72	0
BO	Bolivia	1.6	14.70	0
BR	Brazil	2.2	15.70	0
CN	China	4.6	12.90	1
CY	Cyprus	2.8	12.17	0
DK	Denmark	2.4	15.07	0
EC	Ecuador	0.1	17.59	0
FR	France	0.7	14.50	1
GB	United Kingdom	0.5	14.30	1
GT	Guatemala	3.5	13.24	0
HK	Hong Kong	2.7	11.19	0
HR	Croatia	3.0	14.77	0
HT	Haiti	-2.3	15.67	0
IE	Ireland	2.5	12.78	0
IL	Israel	1.6	13.60	1
IN	India	6.8	12.93	1
IQ	Iraq	1.4	11.35	0
IR	Iran	3.3	11.58	0
IT	Italy	0.7	12.50	0
JM	Jamaica	1.8	16.30	0
JO	Jordan	2.6	13.50	0
KR	South Korea	2.3	10.80	0
LB	Lebanon	0.9	16.82	0
MA	Morocco	3.1	13.86	0
MK	North Macedonia	2.8	13.98	0
MX	Mexico	2.4	14.09	0
NG	Nigeria	3.4	17.00	0
NO	Norway	2.1	14.34	0
NP	Nepal	3.9	9.98	0
NZ	New Zealand	1.0	15.79	0
PG	Papua New Guinea	4.5	13.62	0
PK	Pakistan	2.5	11.40	1
PL	Poland	3.1	15.41	0
PT	Portugal	2.0	15.14	0
RU	Russia	3.2	14.16	1
SE	Sweden	0.6	15.36	0
SV	El Salvador	2.9	12.71	0
TH	Thailand	2.7	9.43	0
TN	Tunisia	1.6	14.61	0
UA	Ukraine	3.2	14.20	0
US	United States	2.7	14.15	1
VN	Vietnam	5.8	10.15	0
ZA	South Africa	1.2	15.29	0

3 Econometric Model

Let

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{X} = \begin{bmatrix} 1 & X_{11} & D_1 \\ 1 & X_{12} & D_2 \\ \vdots & \vdots & \vdots \\ 1 & X_{1n} & D_n \end{bmatrix}.$$

The OLS estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}, \quad (2)$$

where

$$\hat{\beta} = \begin{bmatrix} \hat{a} \\ \hat{b} \\ \hat{c} \end{bmatrix}.$$

The fitted model is

$$\hat{y}_i = 7.2155 - 0.3541X_{1i} + 0.3876D_i. \quad (3)$$

4 Regression Results

Table 2: OLS estimates

Variable	Coefficient	Std. Error	<i>t</i> -Statistic	<i>p</i> -Value
Intercept <i>a</i>	7.2155	1.580	4.567	0.000
<i>X</i> ₁	-0.3541	0.113	-3.138	0.003
<i>D</i> = 1 _{<i>X</i>₂=1}	0.3876	0.548	0.707	0.484

Table 3: Model diagnostics

Statistic	Value
Observations	43
<i>R</i> ²	0.211
Adjusted <i>R</i> ²	0.172
F-statistic	5.358
Prob(F-statistic)	0.00867
Durbin-Watson	2.504

5 Interpretation

The coefficient on *X*₁ is

$$\hat{b} = -0.3541.$$

Thus, holding nuclear-armed status fixed, a one-centimeter increase in average erect penis length is associated with a decrease of approximately 0.3541 units in *y*.

The coefficient on nuclear-armed status is

$$\hat{c} = 0.3876.$$

Thus, holding *X*₁ fixed, nuclear-armed countries have predicted *y*-values approximately 0.3876 units higher than non-nuclear countries. However, this coefficient is not statistically significant at conventional levels.

6 Geopolitical and Warfare-Economic Interpretation

Nuclear-armed status is a strategic variable associated with deterrence, regime capacity, technological capability, military doctrine, and international bargaining power. In a warfare-economics framework, one may view nuclear status as a costly security investment that changes a state's strategic payoff matrix.

Let a state's simplified security payoff be

$$U_i = B_i(D_i) - C_i(D_i) + E_i,$$

where *B*_{*i*}(*D*_{*i*}) is the deterrence benefit, *C*_{*i*}(*D*_{*i*}) is the cost of maintaining nuclear capability, and *E*_{*i*} captures economic and institutional fundamentals. The regression coefficient $\hat{c}$ estimates only a reduced-form conditional mean difference:

$$\hat{c} \neq \frac{\partial U_i}{\partial D_i}.$$

It is therefore not a causal estimate of the effect of nuclear weapons on *y*.

7 Statistical Caution

The model is observational. Hence,

$$\mathbb{E}[\varepsilon_i \mid X_{1i}, D_i] = 0$$

is a strong assumption and may fail because of omitted variables such as income, geography, institutions, education, demographics, colonial history, health systems, conflict exposure, and measurement error.

The estimated relation is therefore best interpreted as

$$\text{Corr}(y_i, X_{1i}, D_i),$$

not as a structural law.

8 Algorithmic Replication

The following pseudo-code describes the estimation procedure.

Table 4: OLS algorithm

Step	Operation
1	Input country-level observations (y_i, X_{1i}, X_{2i}) .
2	Define $D_i = \mathbf{1}_{\{X_{2i}=1\}}$.
3	Construct design matrix $\mathbf{X} = [\mathbf{1}, X_1, D]$.
4	Estimate $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$.
5	Compute fitted values $\hat{\mathbf{y}} = \mathbf{X}\hat{\beta}$.
6	Compute residuals $\hat{\varepsilon}_i = y_i - \hat{y}_i$.
7	Report coefficients, standard errors, t -statistics, p -values, and R^2 .

9 AI, Machine Learning, and Robustness Extensions

A machine-learning robustness workflow may compare OLS with regularized and nonparametric estimators:

$$\hat{f} = \arg \min_f \sum_{i=1}^n (y_i - f(X_{1i}, D_i))^2 + \lambda \Omega(f),$$

where $\Omega(f)$ is a complexity penalty.

Possible models include:

OLS Linear benchmark.

Ridge regression Penalized linear regression with ℓ_2 shrinkage.

LASSO Penalized regression with ℓ_1 shrinkage.

Random forest Nonlinear ensemble estimator.

Bayesian regression Probabilistic coefficient inference.

Causal forest Heterogeneous treatment-effect exploration, if credible identifying assumptions exist.

Because $n = 43$, complex ML models would have high variance. Therefore, interpretability and parsimony are preferable.

10 Vector Graphic

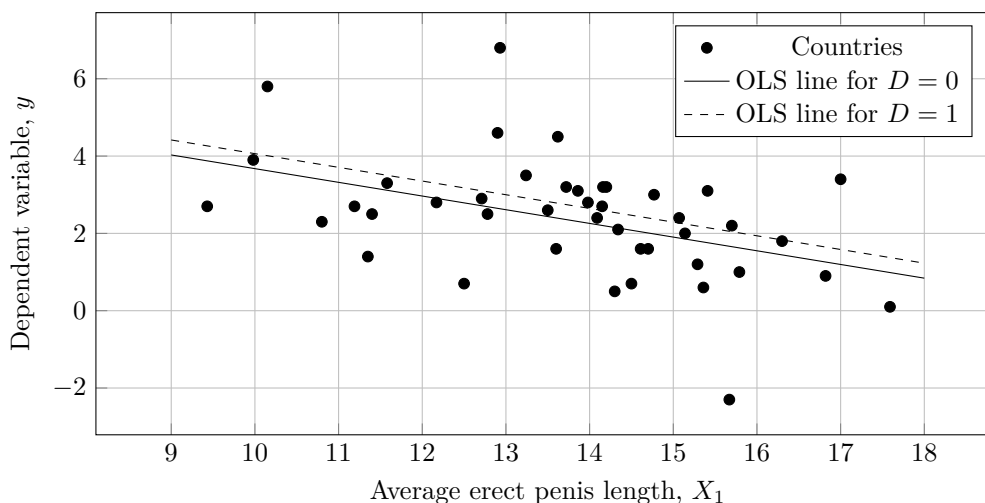


Figure 1: Observed y against X_1 , with fitted conditional regression lines for non-nuclear and nuclear-armed states.

11 Conclusion

The estimated regression is

$$\hat{y}_i = 7.2155 - 0.3541X_{1i} + 0.3876D_i.$$

Within this dataset, X_1 is negatively associated with y , while nuclear-armed status is positively but insignificantly associated with y . The results are reduced-form correlations, not causal effects. A more rigorous structural or causal study would require stronger identification, additional covariates, robustness checks, and careful measurement validation.

References

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Glossary

OLS

Ordinary least squares, a method for estimating linear regression coefficients.

Indicator variable

A binary variable equal to 1 when a condition is true and 0 otherwise.

Nuclear-armed nation

A country possessing nuclear weapons capability.

Reduced-form regression

A regression estimating conditional associations without explicitly modeling all structural mechanisms.

Omitted-variable bias

Bias arising when relevant explanatory variables are excluded from the regression.

Durbin–Watson statistic

A statistic used to detect residual autocorrelation.

Warfare economics

The study of economic incentives, constraints, and resource allocation in conflict and military strategy.

Geopolitics

The study of power, geography, strategy, and international political relations.

The End